Robustness of deep models

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Short-cut learning [3]

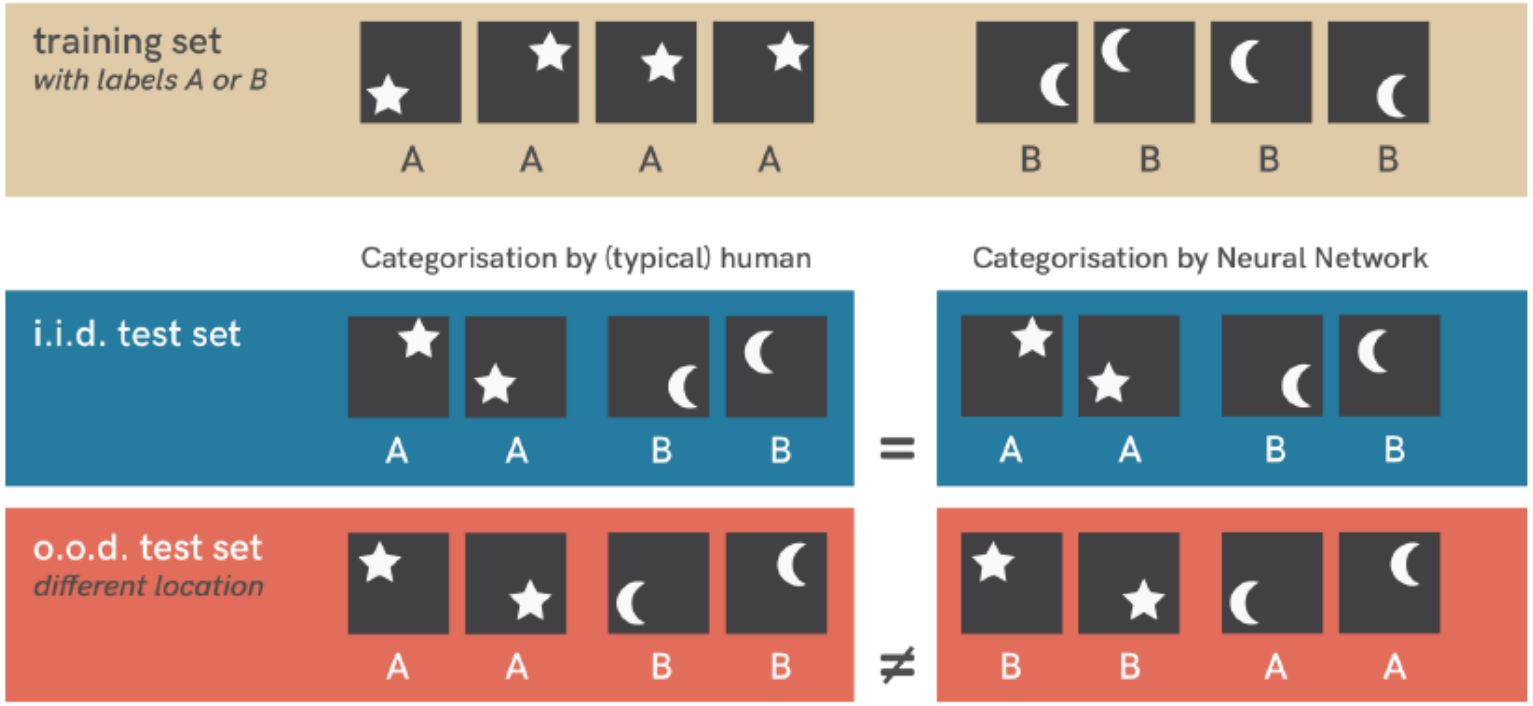
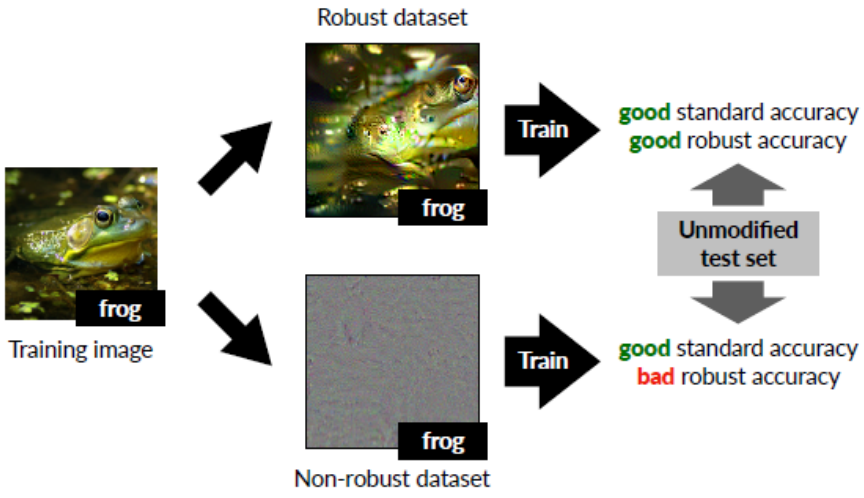
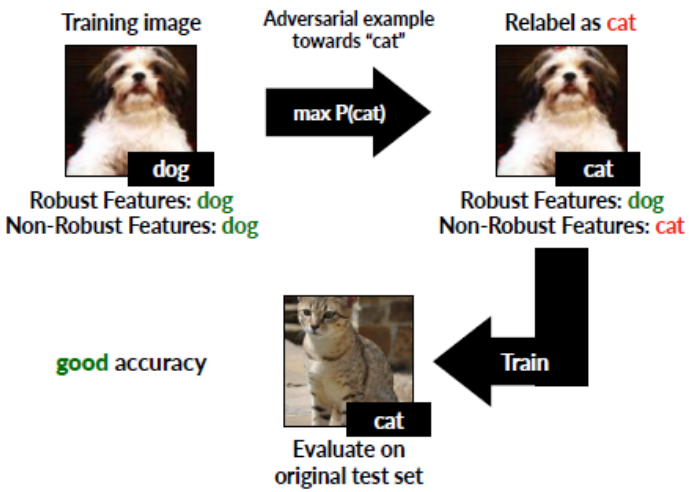


Figure 1: When trained on a simple dataset of stars and moons (top row), a standard neural network (three layers, fully connected) can easily categorise novel similar exemplars (mathematically termed i.i.d. test set). However, testing it on a slightly different dataset (o.o.d. test set, bottom row) reveals a shortcut strategy: The network has learned to associate object location with a category. During training, stars were always shown in the top right or bottom left of an image; moons in the top left or bottom right.

Adversarial examples are not bugs, they are features [5]



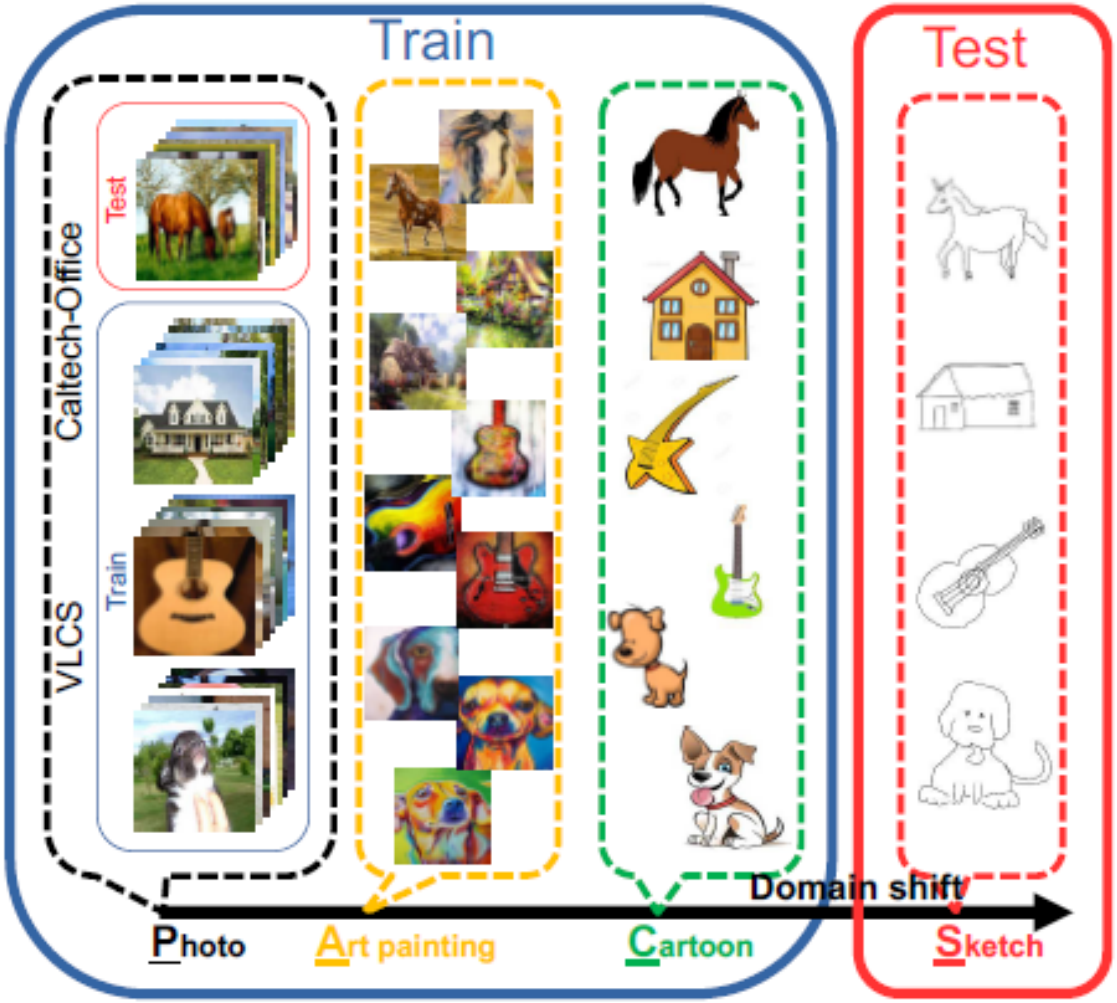
(a)



(b)

Notice: Having high test accuracy does not implies robust learning. Deep models learn features, but features may not be **robust**.

Domain shift



Same same but different :) How can we learn the semantic?

Class-activation mapping

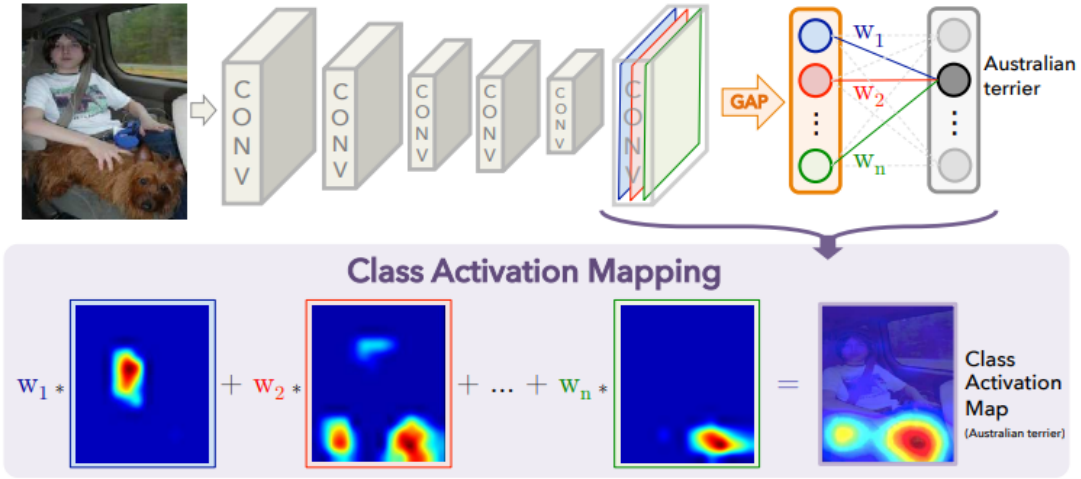


Figure 2: Illustration of class-activation mapping (CAM)

For a given image, let $f_k(x, y)$ represent the activation of unit k in the last convolutional layer at spatial location (x, y). Then, for unit k, the result of performing global average pooling is

$$F_k = \sum_{x,y} f_k(x, y).$$

Class-activation mapping

Figure 3: Illustration of class-activation mapping (CAM)

Thus, for a given class c, the input to the softmax, S_c, is

$$S_c = \sum_k w_k^c F_k$$

where w_k^c is the weight corresponding to class c. Essentially, w_k^c indicates the importance of F_k for class c.

Class-activation mapping

Figure 4: Illustration of class-activation mapping (CAM)

Finally the output of the softmax for class c, P_c, is given by

$$P_c = \frac{\exp(S_c)}{\sum_c \exp(S_c)}.$$

Class-activation mapping

Notice By plugging $F_k = \sum_{x,y} f_k(x,y)$ into the class score, S_c, we obtain

$$S_c = \sum_k w_k^c \sum_{x,y} f_k(x,y) = \sum_{x,y} \sum_k w_k^c f_k(x,y). \tag{1}$$

We define M_c as the class activation map for class c, where each spatial element is given by

$$M_c(x,y) = \sum_k w_k^c f_k(x,y). \tag{2}$$

Thus, $S_c = \sum_{x,y} M_c(x,y)$, and hence $M_c(x,y)$ directly indicates the importance of the activation at spatial grid (x,y) leading to the classification of an image to class c.

GradCam [6]

- **Step 1 — Forward Pass**
  Feed the input image $\mathbf{x}$ through the network. Let y^c denote the (pre-softmax) score for target class c. Let

  $$A^k \in \mathbb{R}^{H \times W}, \quad k = 1, \ldots, K$$

  denote the K feature maps produced by the final convolutional layer, where H and W are the spatial dimensions.
- **Step 2 — Compute Gradients**
  Backpropagate the gradient of y^c with respect to each feature map A^k:

  $$\frac{\partial y^c}{\partial A^k_{ij}}, \quad i \in \{1, \ldots, H\},\ j \in \{1, \ldots, W\}.$$

  This gradient quantifies how much a change at spatial location (i, j) in feature map k would affect the class score.

GradCam [6]

- **Step 3 — Global Average Pooling of Gradients**
  Compute a scalar importance weight α_k^c for each feature map by globally average-pooling the gradients:

$$\boxed{\alpha_k^c = \frac{1}{Z} \sum_{i=1}^{H} \sum_{j=1}^{W} \frac{\partial y^c}{\partial A_{ij}^k}}$$

  where $Z = H \times W$ is a normalisation constant. Intuitively, α_k^c captures the **global importance** of feature map k for class c.
- **Step 4 — Weighted Combination and ReLU**
  Form the GradCAM localisation map $L^c \in \mathbb{R}^{H \times W}$ as a weighted combination of the feature maps, followed by a ReLU:

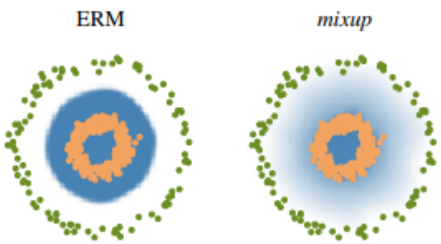
$$\boxed{L_{\text{GradCAM}}^c = \text{ReLU}\left(\sum_{k=1}^{K} \alpha_k^c \, A^k \right)}$$

  The ReLU retains only those activations that have a *positive* influence on the target class.

GradCam [6]

```
# y1, y2 should be one-hot vectors
for (x1, y1), (x2, y2) in zip(loader1, loader2):
    lam = numpy.random.beta(alpha, alpha)
    x = Variable(lam * x1 + (1. - lam) * x2)
    y = Variable(lam * y1 + (1. - lam) * y2)
    optimizer.zero_grad()
    loss(net(x), y).backward()
    optimizer.step()
```

(a) One epoch of *mixup* training in PyTorch.



(b) Effect of *mixup* ($\alpha = 1$) on a toy problem. Green: Class 0. Orange: Class 1. Blue shading indicates $p(y = 1|x)$.

Recall: Previously we have discussed, for training DNNs, we use ERM, since we do not know the true data distribution, which may result in overfitting.

Data Augmentation: MixUp [8]

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# y1, y2 should be one-hot vectors
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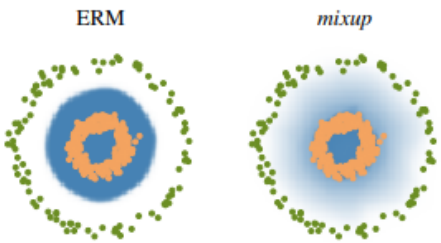

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MixUp constructs virtual training examples

$$\tilde{x} = \lambda x_i + (1 - \lambda) x_j, \quad \text{where } x_i, x_j \text{ are raw input vectors} \tag{1}$$

$$\tilde{y} = \lambda y_i + (1 - \lambda) y_j, \quad \text{where } y_i, y_j \text{ are one-hot label encodings} \tag{2}$$

(x_i, y_i) and (x_j, y_j) are two examples drawn at random from our training data, and $\lambda \in [0, 1]$.

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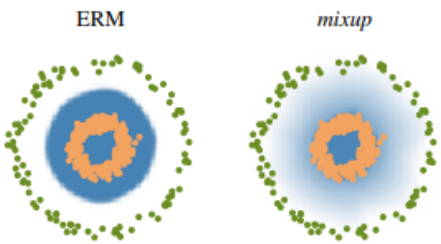

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Data Augmentation: CutMix [7]

	ResNet-50	Mixup [48]	Cutout [3]	CutMix
Image				
Label	Dog 1.0	Dog 0.5 Cat 0.5	Dog 1.0	Dog 0.6 Cat 0.4
ImageNet Cls (%)	76.3 (+0.0)	77.4 (+1.1)	77.1 (+0.8)	**78.6** **(+2.3)**
ImageNet Loc (%)	46.3 (+0.0)	45.8 (-0.5)	46.7 (+0.4)	**47.3** **(+1.0)**
Pascal VOC Det (mAP)	75.6 (+0.0)	73.9 (-1.7)	75.1 (-0.5)	**76.7** **(+1.1)**

Figure 5: Comparison of different augmentation methods.

Data Augmentation: CutMix [7]

- Let $x \in \mathbb{R}^{W \times H \times C}$ and y denote a training image and its label, respectively.
- CutMix generate a new training sample $(\tilde{x}, \tilde{y})$ by combining two training samples (x_A, y_A) and (x_B, y_B).
- The generated training sample $(\tilde{x}, \tilde{y})$ is used to train the model with its original loss function.
- We define the combining operation as

$$\tilde{x} = \mathbf{M} \odot x_A + (\mathbf{1} - \mathbf{M}) \odot x_B \tag{3}$$

$$\tilde{y} = \lambda y_A + (1 - \lambda) y_B \tag{4}$$

  where $\mathbf{M} \in \{0, 1\}^{W \times H}$ denotes a binary mask indicating where to drop out and fill in from two images, $\mathbf{1}$ is a binary mask filled with ones, and $\odot$ is element-wise multiplication.
- λ is sampled from the beta distribution $\text{Beta}(\alpha, \alpha)$.

Data Augmentation: CutMix [7]

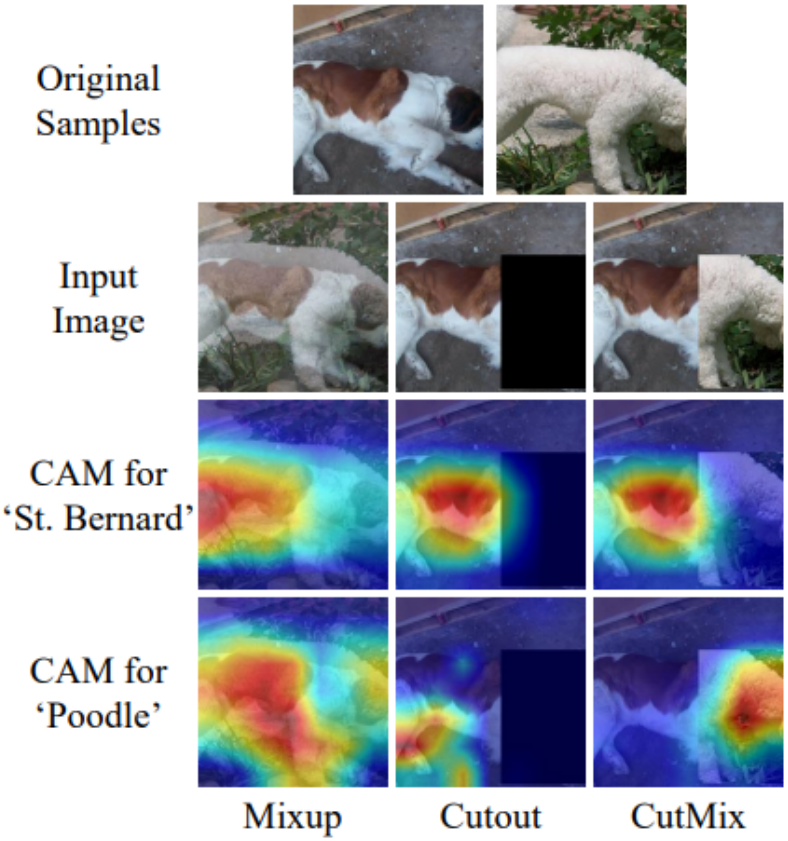

Figure 6: Illustration of CAM for different augmentations

Notice: CutMix helps network to focus on class-specific features.

Data Augmenttaion : AutoAugment [1]

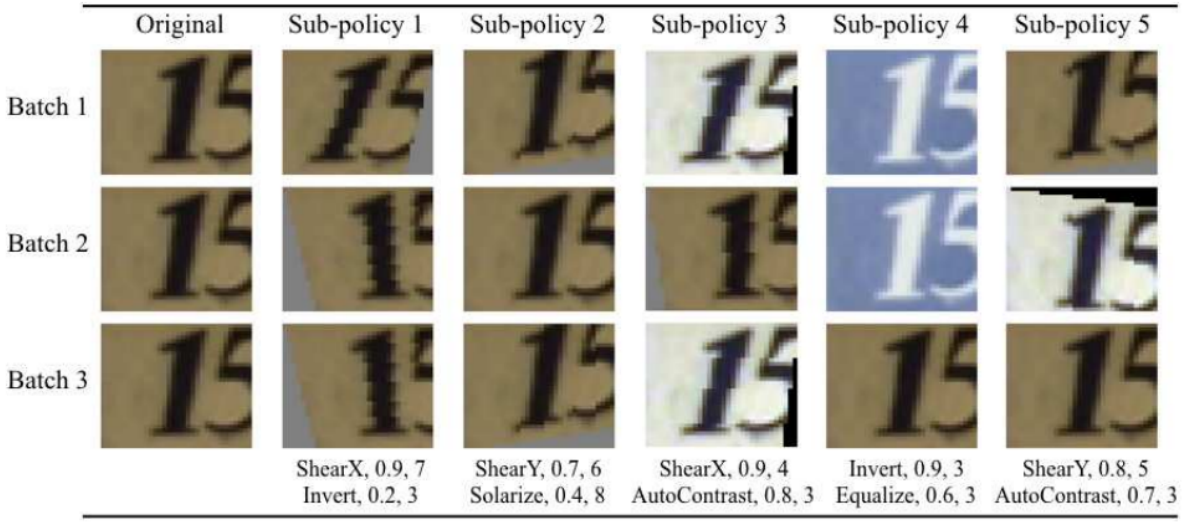

Figure 7: Illustration of sub-policies.

Search space: A policy consists of 5 sub-policies with each sub-policy consisting of two image operations to be applied in sequence. Additionally, each operation is also associated with two hyperparameters: 1) the probability of applying the operation, and 2) the magnitude of the operation.

Data Augmentation AutoAugment [1]

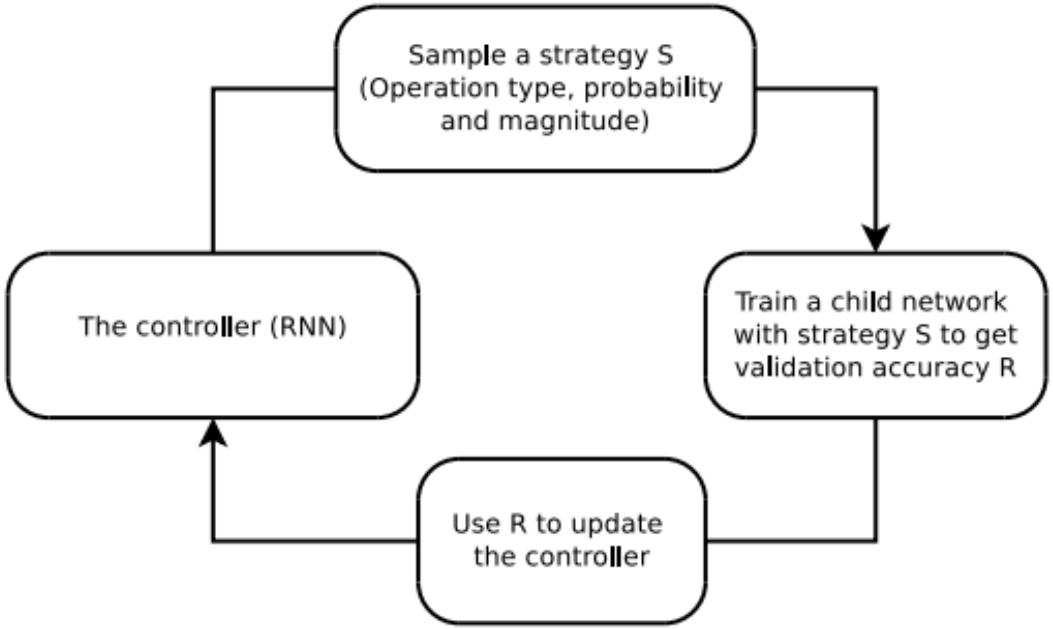

Figure 8: Illustration of sequence of transformations in Randaugment.

- Use a search method (e.g., Reinforcement Learning) to seek better data augmentation policies.
- A controller RNN predicts an augmentation policy from the search space.
- A child network with a fixed architecture is trained to convergence

Data Augmentation: Random augmentation [2]

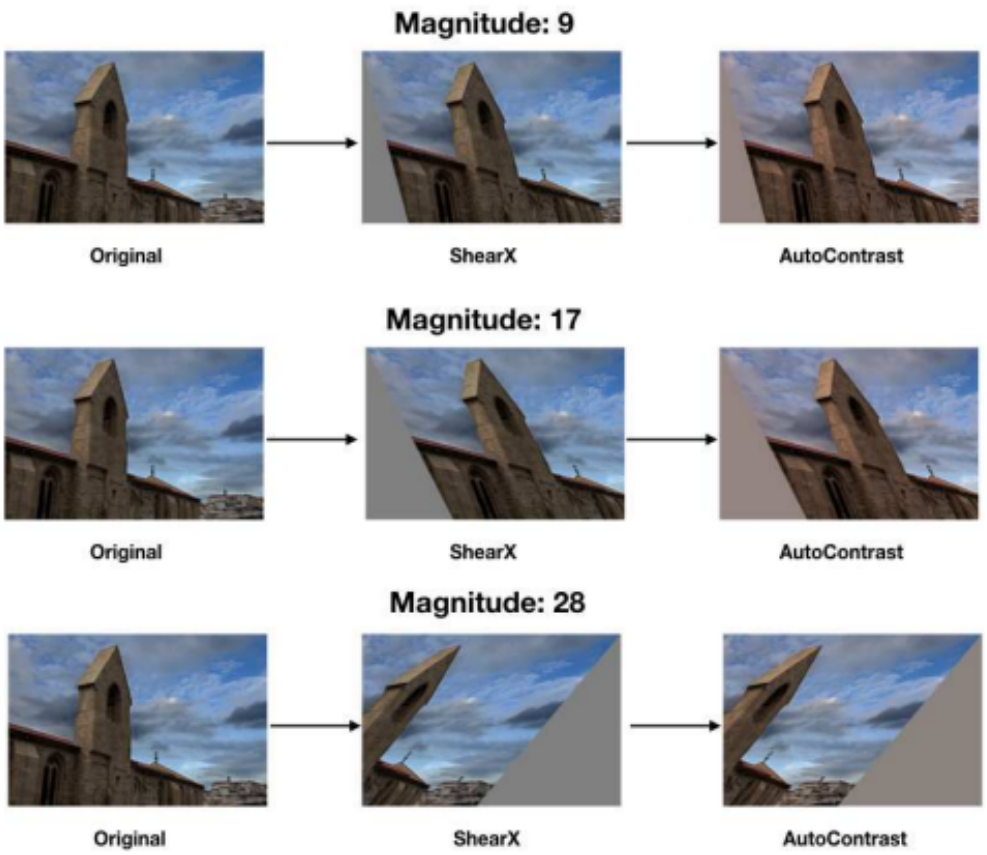

Figure 9: Illustration of sequence of transformations in Randaugment.

Data Augmentation: Random augmentation [2]

```
transforms = [
'Identity', 'AutoContrast', 'Equalize', 'Rotate',
     'Solarize', 'Color',
'Posterize', 'Contrast', 'Brightness', 'Sharpness',
     'ShearX', 'ShearY',
'TranslateX', 'TranslateY']

def randaugment(N, M):
"""Generate a set of distortions.

  Args:
    N: Number of augmentation transformations to apply
         sequentially.
    M: Magnitude for all the transformations.
"""

 sampled_ops = np.random.choice(transforms, N)
 return [(op, M) for op in sampled_ops]
```

Figure 10: Illustration of the implementation.

Notice: Randaugment has two parameters; number of transformations (N) and magnitude of transformation (M).

Data Augmentation: AugMix [4]

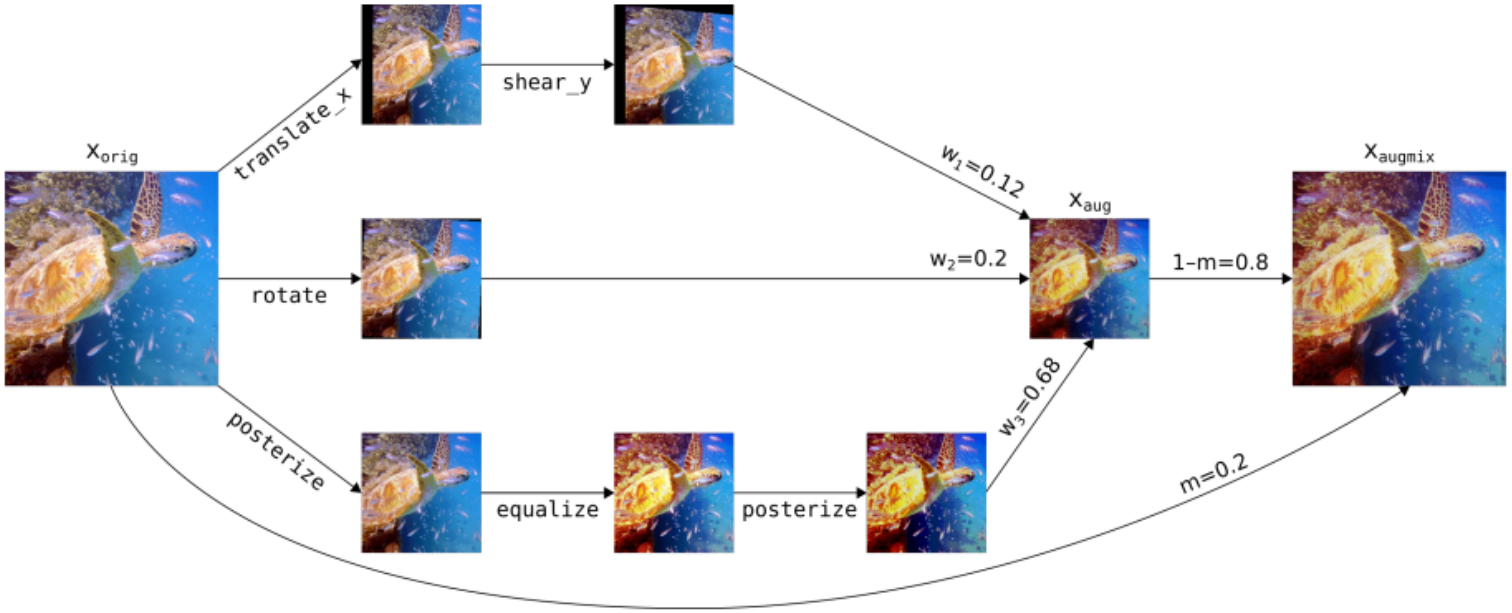

Figure 11: Illustration of generating augmented sample X_{augmix}

Notice: AugMix first combines different augmented images (with different sequence depth of transformations), then combines it with original image.

Data Augmentation: AugMix [4]

Algorithm AUGMIX Pseudocode

Input: Model $\hat{p}$, Classification Loss $\mathcal{L}$, Image x_{orig}, Operations $\mathcal{O} = \{\text{rotate}, \ldots, \text{posterize}\}$
function AugmentAndMix(x_{orig}, $k = 3, \alpha = 1$)
  Fill x_{aug} with zeros
  Sample mixing weights $(w_1, w_2, \ldots, w_k) \sim \text{Dirichlet}(\alpha, \alpha, \ldots, \alpha)$
  for $i = 1, \ldots, k$ **do**
    Sample operations $\text{op}_1, \text{op}_2, \text{op}_3 \sim \mathcal{O}$
    Compose operations with varying depth $\text{op}_{12} = \text{op}_2 \circ \text{op}_1$ and $\text{op}_{123} = \text{op}_3 \circ \text{op}_2 \circ \text{op}_1$
    Sample uniformly from one of these operations chain $\sim \{\text{op}_1, \text{op}_{12}, \text{op}_{123}\}$
    $x_{\text{aug}} \mathrel{+}= w_i \cdot \text{chain}(x_{\text{orig}})$ ▷ *Addition is elementwise*
  end for
  Sample weight $m \sim \text{Beta}(\alpha, \alpha)$
  Interpolate with rule $x_{\text{augmix}} = m x_{\text{orig}} + (1 - m) x_{\text{aug}}$
  return x_{augmix}
end function
$x_{\text{augmix1}} = \text{AugmentAndMix}(x_{\text{orig}})$ ▷ $x_{augmix1}$ *is stochastically generated*
$x_{\text{augmix2}} = \text{AugmentAndMix}(x_{\text{orig}})$ ▷ $x_{augmix1} \neq x_{augmix2}$
Loss Output: $\mathcal{L}(\hat{p}(y \mid x_{\text{orig}}), y) + \lambda\, \text{Jensen-Shannon}(\hat{p}(y \mid x_{\text{orig}}); \hat{p}(y|x_{\text{augmix1}}); \hat{p}(y|x_{\text{augmix2}}))$

Figure 12: AugMix Algorithm.

Data Augmentation: AugMix [4]

- Minimize the Jensen-Shannon divergence among the posterior distributions of the original sample x_{orig} and its augmented variants.
- For $p_{\text{orig}} = \hat{p}(y \mid x_{\text{orig}})$, $p_{\text{augmix1}} = \hat{p}(y \mid x_{\text{augmix1}})$, $p_{\text{augmix2}} = \hat{p}(y \mid x_{\text{augmix2}})$, we replace the original loss $\mathcal{L}$ with:

$$\mathcal{L}(p_{\text{orig}}, y) + \lambda \, \mathrm{JS}(p_{\text{orig}};\, p_{\text{augmix1}};\, p_{\text{augmix2}}) \tag{5}$$

Interpretation

Imagine sampling from one of the three distributions $p_{\text{orig}}, p_{\text{augmix1}}, p_{\text{augmix2}}$. The Jensen-Shannon divergence measures the average information that the sample reveals about *which* distribution it was drawn from. Minimising it forces the model to be invariant to augmentation.

Data Augmentation: AugMix [4]

- Computing JS via the average mixture $M = \dfrac{p_{\text{orig}} + p_{\text{augmix1}} + p_{\text{augmix2}}}{3}$:

$$\begin{aligned}&\mathrm{JS}(p_{\text{orig}};\, p_{\text{augmix1}};\, p_{\text{augmix2}})\\ &= \frac{1}{3}\Big(\mathrm{KL}[p_{\text{orig}} \| M] + \mathrm{KL}[p_{\text{augmix1}} \| M] + \mathrm{KL}[p_{\text{augmix2}} \| M]\Big)\end{aligned} \tag{6}$$

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In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2019.

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